

Assignment # 02: Practicing Control-Flow (Decision-Making Structures)

Objective:

To help you understand and apply control-flow in Python Programming.

- Use **if**, **if-else** and **if-elif-else** statements to make decisions in a program.
- Develop logical thinking.

Decision-Making Structures

1. Write a program that checks if a given number is positive, negative or zero.
2. Write a program to find the larger of the two numbers.
3. Write a program to find the largest of the three numbers.
4. Write a program to check if the given string "Mass" is present in the string "Saylani Mass IT". If the string is present then display the message "string found".
5. Write a program to take user's age as input and display whether he is minor, adult or senior citizen:
 - a. Minor age is less than 18.
 - b. Adult age is greater than 18 and less than 60
 - c. Senior citizen age is greater than 60
6. Write a program to take integer as input and check if it is even or odd.
7. Write a program to take two numbers and an operator (/ , x , + , -) as input and perform the corresponding operation.
8. Write a program to take number as input and check if it lies in the range i.e. 20(inclusive) – 40 (exclusive)
9. Write a program that asks for an integer number and checks if it is divisible by 2, 3, or both.
10. Write a program to take user's score as input and determine if they pass or fail (pass if score is above 60, otherwise fails.)
11. Write a program that takes a temperature in Celsius and checks if it is freezing, moderate or hot.
 - a. Freezing temperature is below 0.
 - b. Moderate temperature is greater than 0 and less than 26.
 - c. Hot temperature is above 26.
12. Write a program to input salary, if salary is greater than Rs. 600000 then deduct 5% tax and calculate the Gross Salary.
13. Write a program to input salary and deduct Tax and calculate gross salary
 - a. if salary is greater than 2500000 ---> 15% Tax Deduction
 - b. if salary is between 1500000 - 2500000 ---> 10% Tax Deduction
 - c. if salary is between 500000 - 1500000 ---> 5% Tax Deduction
 - d. if salary is less than 500000 ---> 0% Tax Deduction